Where is Rhyming? Phonetic Representations are Front-Loaded in Transformer Language Models

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Abstract

Language models can identify rhyming words despite never hearing speech, raising the question: where is this phonetic knowledge encoded? We investigate the location and geometry of phonetic representations in GPT-2 MEDIUM (24 layers, 1024 dimensions) using 13,105 single-token English words with IPA transcriptions from WIKIPRON. We train per-phoneme linear probes at every layer of the residual stream and find that phonetic information is *front-loaded*: probe accuracy peaks at layer 0 (macro F1 = 0.41) and does not improve in deeper layers, contrasting sharply with semantic features that build through the network. We then analyze phoneme direction vectors—mean activation differences for words containing versus not containing each phoneme—and find that each phoneme has a distinct direction (mean pairwise $|\text{cosine similarity}| = 0.13$, $5.3\times$ above random), but these directions are not orthogonal subspaces; they become increasingly aligned in deeper layers as semantic representations develop. Finally, we test context-dependent rhyming with GPT-4.1 on 12 heteronyms and find a systematic primary-pronunciation bias: 92% accuracy on primary but only 58% on secondary pronunciations. Together, these results show that rhyming knowledge lives primarily in the token embedding table, is carried but not enriched through the network, and that resolving context-dependent pronunciation remains an open challenge for current models.

1 Introduction

Language models trained exclusively on text can tell you that "cat" rhymes with "bat" but not with "cut"—despite having never heard either word spoken. This ability is striking: rhyming is fundamentally a *phonetic* relationship, yet these models receive no auditory signal, no phonemic annotation, and no explicit pronunciation data during training. The co-occurrence statistics of written language are apparently sufficient for models to recover aspects of the sound system. But *where* in the model does this phonetic knowledge reside?

Recent work by McLaughlin et al. [2025] provided a first answer for LLAMA-3.2-1B: a "phoneme mover head" (H13L12) organizes rhyming behavior, and linear probes on the embedding layer recover IPA phonemes with approximately 96% accuracy. However, several questions remain open. First, does phonetic information *build* through the network the way syntactic and semantic features do [Radford et al., 2019], or is it established early and simply maintained? Second, does each IPA phoneme correspond to a distinct, roughly orthogonal subspace—or do phonemes share representational structure in more complex ways? Third, how do models handle *context-dependent* pronunciation, as in heteronyms like "read" (/ri:d/ vs. /rɛd/)?

We address these questions through three experiments on GPT-2 MEDIUM (355M parameters, 24 layers). Using 13,105 single-token English words with IPA transcriptions from WIKIPRON, we train per-phoneme Ridge classifiers at every layer of the residual stream and find that phonetic information is *front-loaded*: it peaks at layer 0 (macro F1 = 0.41) with no improvement in deeper layers (section 4.1). We compute phoneme direction vectors and show that each phoneme has a distinct direction ($5.3\times$ above random baseline) but these directions are not orthogonal—they become

more aligned in deeper layers (section 4.2). We test GPT-4.1 on heteronym rhyming and uncover a systematic primary-pronunciation bias: 92% accuracy on primary but only 58% on secondary pronunciations (section 4.3).

Our contributions are:

- We conduct the first systematic layer-by-layer phoneme probing across all 25 layers of a transformer language model, showing that phonetic information is front-loaded and does not build through the network.
- We analyze phoneme subspace geometry using direction vectors, finding structured but non-orthogonal phoneme representations that compress in deeper layers.
- We introduce a heteronym rhyming test that reveals a primary-pronunciation bias in modern language models, demonstrating that context-dependent phonetic processing remains an open challenge.

2 Related Work

Phonetic representations in language models. The question of whether text-only models learn phonetic information has attracted growing interest. McLaughlin et al. [2025] provided the most rigorous mechanistic study to date, identifying a "phoneme mover head" in LLAMA-3.2-1B that promotes phonetically similar tokens during rhyme generation, with linear probes achieving approximately 96% accuracy on embedding-layer phoneme prediction. Their work demonstrated that phonetic information is linearly accessible in the embedding layer, but did not systematically track how this information evolves across all layers of the residual stream. We extend their analysis by probing every layer of GPT-2 MEDIUM, revealing that phonetic information is front-loaded rather than progressively built. Bunzeck et al. [2024] compared grapheme-based and phoneme-based character-level language models on phonetic tasks, finding that even small models achieve 88–92% rhyme prediction accuracy, with grapheme models surprisingly outperforming phoneme models. Earlier work on character embeddings [Elhage et al., 2022] established that sub-word representations carry perceptual properties including sound-related features.

Phonetic word embeddings. Zouhar et al. [2024] developed a standardized evaluation suite for phonetic word embeddings, including rhyme detection as a benchmark task. Their work uses count-based, autoencoder, and contrastive learning methods with articulatory features from PanPhon, providing complementary evaluation methodology for measuring phonetic content in learned representations. Our direction-vector analysis relates to this line of work: rather than training specialized phonetic embeddings, we analyze the implicit phonetic structure in general-purpose language model representations.

Rhyme in neural text generation. Several systems have addressed rhyme explicitly in the context of poetry or song generation. Lau et al. [2018] proposed a joint neural model of poetic language, meter, and rhyme for sonnet generation. Jhamtani et al. [2019] used structured adversarial training to enforce rhyming constraints without manual phonetic features. Xue et al. [2021] built a transformer-based rap generation system with explicit rhyme and rhythm modeling. Haider and Kuhn [2018] developed Siamese recurrent networks for supervised rhyme detection. These systems treat rhyme as an explicit constraint to be enforced; our work instead asks where rhyme knowledge *already exists* within standard pretrained models.

Mechanistic interpretability of linguistic features. Our probing methodology follows the broader tradition of using linear probes to identify linguistic features in neural network representations [Radford et al., 2019]. While most probing studies find that syntactic and semantic features build through the network—with syntax peaking in middle layers and semantics in later layers—we find a qualitatively different pattern for phonetic features: they are strongest at the embedding boundary and gradually degrade. This contributes to a more complete picture of how different levels of linguistic structure are distributed across transformer layers.

3 Methodology

3.1 Dataset Construction

We construct a dataset of English words with IPA transcriptions by intersecting the WIKIPRON US English broad transcription database [Lee et al., 2020] with the set of words that encode as a single token under the GPT-2 tokenizer. This single-token constraint ensures a clean mapping between token representations and phonemic content: each word's embedding directly corresponds to its complete pronunciation without BPE segmentation artifacts.

The resulting dataset contains 13,105 words spanning 77 unique IPA phonemes. Of these, 4,607 words (35%) appear in the OXFORD-5000 common word list, confirming broad coverage of everyday English. We use the first transcription listed in WIKIPRON for words with multiple pronunciation entries. Each word is labeled with a multi-hot vector indicating which IPA phonemes it contains (e.g., "water" maps to the set of its constituent IPA phonemes). We split the data 80/20 into train (10,484) and test (2,621) sets with random seed 42.

3.2 Experiment 1: Layer-Wise Phoneme Probing

Where does phonetic information live across layers? For each of the 25 representation sites in GPT-2 MEDIUM—the token embedding layer plus 24 transformer layer outputs—we extract the residual stream activation $\boldsymbol{h}_\ell \in \mathbb{R}^{1024}$ for each word. We extract activations using TransformerLens [Nanda and Bloom, 2022], processing words in batches of 256 on a single NVIDIA RTX A6000 GPU.

At each layer ℓ, we train 77 independent Ridge classifiers (one per IPA phoneme p), each predicting whether phoneme p is present in the word from the activation vector $\boldsymbol{h}_\ell$. We use Ridge regression ($\alpha = 1.0$) with StandardScaler normalization rather than logistic regression: preliminary experiments showed logistic regression required approximately 10 minutes per layer, while Ridge achieved comparable quality in approximately 11 seconds per layer, enabling a full 25-layer sweep.

We evaluate with macro F1 (averaging across all 77 phonemes equally) and micro F1 (weighting by phoneme frequency). Macro F1 is our primary metric as it better captures performance on rare phonemes.

3.3 Experiment 2: Phoneme Subspace Geometry

Does each IPA sound have a distinct direction in representation space? For each phoneme p with at least 30 occurrences, we compute its *direction vector* as the difference between the mean activation of words containing p and words not containing p:

$$\boldsymbol{d}_p = \frac{1}{|S_p|} \sum_{\boldsymbol{x} \in S_p} \boldsymbol{x} - \frac{1}{|S_p^c|} \sum_{\boldsymbol{x} \in S_p^c} \boldsymbol{x} \tag{1}$$

where $S_p = \{\boldsymbol{x} : p \in \text{phonemes}(\boldsymbol{x})\}$ is the set of activations for words containing phoneme p. We compute these directions at every layer and analyze them via:

- **Pairwise cosine similarity.** We compute the 49×49 matrix of absolute cosine similarities between all phoneme direction vectors and compare the mean to the expected value for random unit vectors in $\mathbb{R}^{1024}$ (≈ 0.025).
- **Clustering analysis.** We apply hierarchical clustering (Ward's method) to the direction vectors and compute silhouette scores for groupings by vowel/consonant distinction and by manner of articulation.
- **PCA visualization.** We project all phoneme directions, vowels only, and consonants only onto their top principal components.
- **Cross-layer comparison.** We track how the mean pairwise similarity and clustering quality evolve from the embedding layer through the final transformer layer.

We use direction vectors rather than Ridge regression weights because the probe weights exhibited near-collinear structure—an artifact of L2 regularization—while mean-difference directions directly measure how each phoneme is encoded in the representation space.

Table 1: Phoneme probing accuracy across layers of GPT-2 MEDIUM. Macro F1 averages equally across all 77 phonemes; micro F1 weights by frequency. Phonetic information peaks at layer 0 and does not improve in deeper layers. Best results in **bold**.

Layer	Macro F1	Micro F1	Exact Match
Embedding	0.395	**0.517**	0.008
Layer 0	**0.413**	0.512	0.007
Layer 1	0.385	0.524	0.009
Layer 5	0.375	0.521	0.009
Layer 11	0.376	0.522	0.009
Layer 17	0.376	0.522	0.009
Layer 23	0.367	0.519	**0.010**

3.4 Experiment 3: Context-Dependent Rhyming

Can models resolve pronunciation when it depends on context? We test whether rhyming knowledge extends beyond static lexical representations by querying GPT-4.1 on heteronyms: words whose pronunciation changes with context (e.g., "read" as /ri:d/ in "I will read" vs. /rɛd/ in "I have read").

We select 12 common English heteronyms (read, lead, wind, tear, bow, bass, dove, minute, close, live, refuse, desert) and construct sentence pairs that establish each pronunciation through syntactic or semantic context. For each context, we present a forced-choice rhyming question with two candidate words (one matching each pronunciation) and record the model's selection. We also include 20 standard (non-heteronym) rhyming pairs as controls: 15 easy pairs and 5 orthographically tricky pairs (e.g., "cough" and "off"). All queries use temperature 0 for deterministic outputs.

4 Results

4.1 Phonetic Information is Front-Loaded

Table 1 shows phoneme probing accuracy across layers. The key finding is the *flatness* of the probing curve: macro F1 peaks at layer 0 (0.413), then plateaus around 0.375 for layers 2–22, with a slight decline to 0.367 at the final layer. Figure 1 shows this pattern across all layers.

This is qualitatively different from how syntactic and semantic features behave in transformer language models, where accuracy typically rises through early and middle layers. Phonetic information is essentially "read off" from the embedding table and its first processing step, then maintained—but not enriched—through the rest of the network.

Performance varies substantially across phonemes. Figure 2 shows per-phoneme F1 scores at the best layer. The most accurately predicted phonemes are /N/ (F1 = 0.80), /I/ (0.70), /@/ (0.69), /z/ (0.67), /n/ (0.60), /t/ (0.60), and /k/ (0.60). However, 47 of 77 phonemes have F1 = 0, almost entirely rare phonemes (aspirated variants, syllabic consonants, infrequent diphthongs) with too few examples in our single-token word set. Of the 49 phonemes with $\geq$30 occurrences, 30 achieve F1 > 0.1.

4.2 Phoneme Directions: Structured but Not Orthogonal

Table 2 summarizes the geometry of phoneme direction vectors. At the embedding layer, the mean pairwise absolute cosine similarity between phoneme directions is 0.133—5.3$\times$ higher than the 0.025 expected for random unit vectors in $\mathbb{R}^{1024}$. This confirms that phonemes have distinct but non-random directions: they share significant representational structure rather than occupying independent subspaces.

Figure 3 shows how this alignment evolves across layers. Phoneme directions become *more* aligned in deeper layers ($0.133 \rightarrow 0.175 \rightarrow 0.270 \rightarrow 0.304$), indicating that the network progressively compresses phonetic variation as it builds semantic representations. The phonetic manifold contracts rather than expanding through the network.

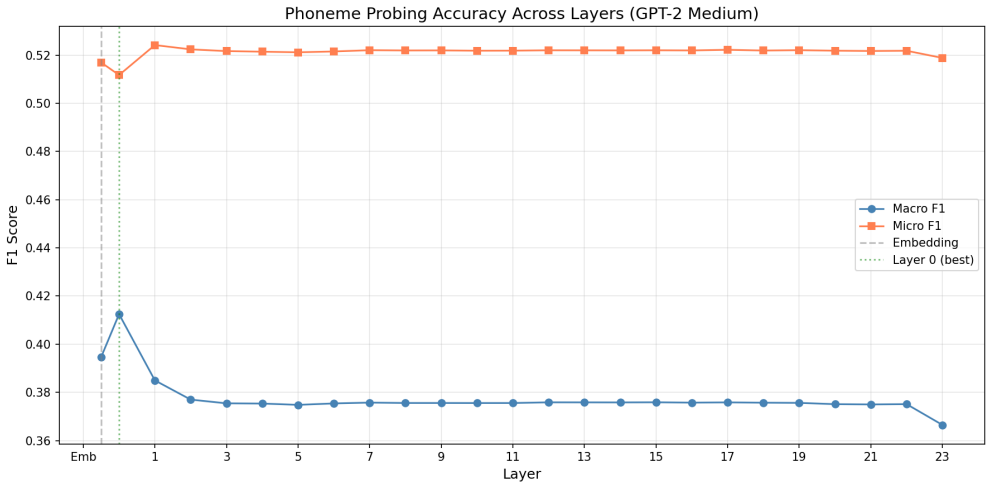


Figure 1: Macro and micro F1 across all 25 layers of GPT-2 MEDIUM. Phonetic probe accuracy peaks at layer 0 and remains flat or slightly declining through deeper layers, indicating phonetic information is established early and not built up through the network.

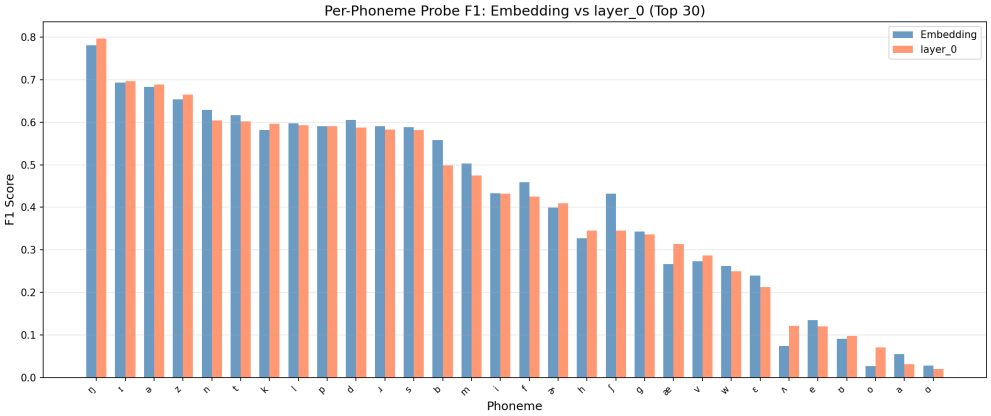


Figure 2: Per-phoneme F1 scores at the best probing layer (layer 0). Common phonemes like N, I, and @ are predicted most accurately, while rare phonemes (aspirated variants, syllabic consonants) have near-zero F1.

Articulatory clustering is absent. Silhouette scores for grouping phoneme directions by vowel/-consonant (-0.032) and manner of articulation (-0.102) are consistently negative (table 2), indicating that phoneme directions in GPT-2 MEDIUM do *not* organize by traditional articulatory categories. This diverges from McLaughlin et al. [2025], who found organized phonetic structure in the result vectors of phoneme mover heads in LLAMA-3.2-1B. The distinction is important: articulatory clustering may emerge in specialized attention head outputs rather than in the general residual stream.

PCA structure. Figure 4 shows PCA projections of phoneme direction vectors at the embedding layer. The top five principal components explain 13.1%, 8.7%, 8.4%, 7.0%, and 6.2% of the variance respectively. While vowels and consonants do not form clean clusters, we observe some local structure: nasal consonants group together, and front vowels separate from back vowels along certain axes.

Table 2: Phoneme direction geometry across layers. Mean |cos sim| measures alignment between phoneme direction vectors; higher values indicate less orthogonal (more overlapping) representations. The expected value for random directions in $\mathbb{R}^{1024}$ is 0.025. Silhouette scores test whether phoneme directions cluster by articulatory categories; negative values indicate no clustering.

Layer	Mean |cos|	Random	Sil. (V/C)	Sil. (Manner)
Embedding	**0.133**	0.025	−0.032	−0.102
Layer 0	0.175	0.025	−0.030	−0.096
Layer 11	0.270	0.025	−0.037	−0.131
Layer 23	0.304	0.025	−0.039	−0.160

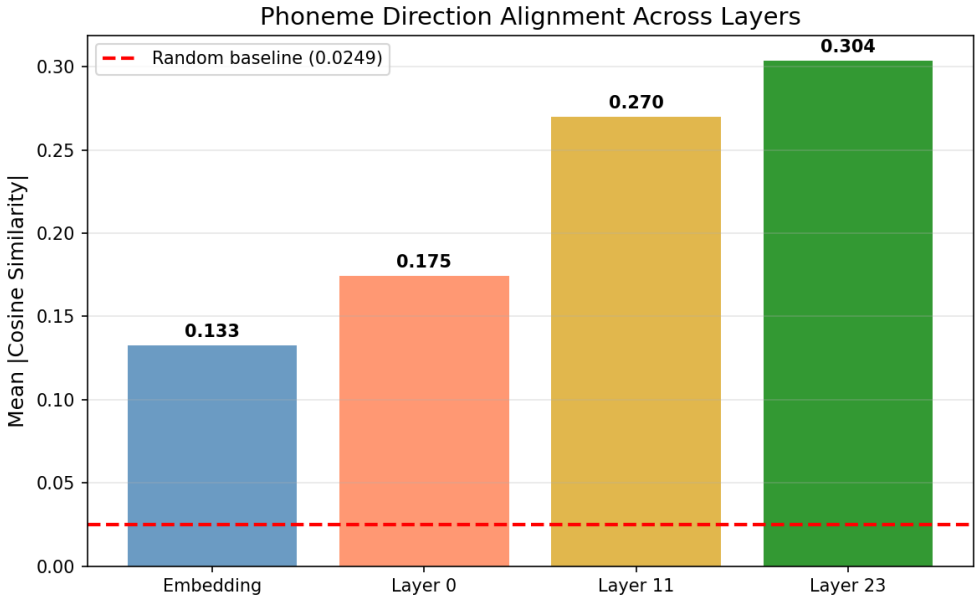


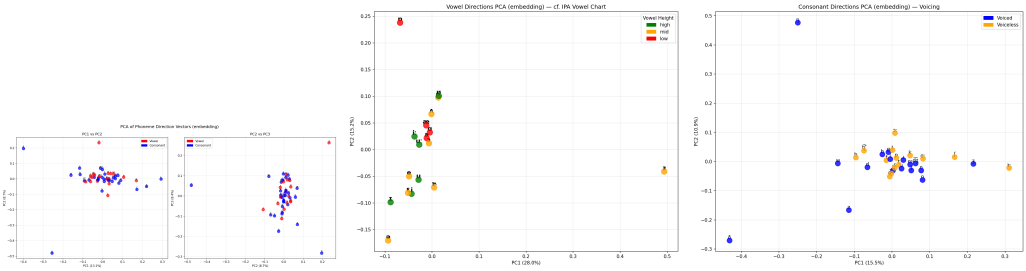
Figure 3: Mean absolute cosine similarity between phoneme direction vectors across layers. Phoneme directions become increasingly aligned (less orthogonal) in deeper layers, suggesting phonetic structure is compressed as semantic representations develop. The dashed line shows the expected value for random vectors.

4.3 Context-Dependent Rhyming Reveals Primary-Pronunciation Bias

Table 3 shows rhyming accuracy for GPT-4.1. On standard rhyming pairs—including orthographically tricky pairs like "cough" and "off"—the model achieves 100% accuracy (20/20). However, heteronyms reveal a systematic asymmetry: 92% accuracy on primary pronunciations versus only 58% on secondary pronunciations.

The failure pattern is informative. GPT-4.1 defaults to the primary pronunciation for "wind" (→ "pinned" not "find"), "bow" (→ "cow" not "show"), "bass" (→ "class" not "face"), "minute" (→ "in it" not "cute"), and "live" (→ "give" not "five"), even when sentence context clearly indicates the secondary pronunciation. It correctly handles context for "read," "lead," "tear," "dove," "close," and "refuse."

This asymmetry is consistent with our probing results: if phonetic information is primarily stored in the static embedding, the model will default to whichever pronunciation dominates in training data. Overriding this default requires contextual processing that the model performs imperfectly.



(a) All phonemes (b) Vowels only (c) Consonants only

Figure 4: PCA projections of phoneme direction vectors at the embedding layer. *(a)* All 49 phonemes with ≥30 occurrences. *(b)* Vowels show partial separation by backness. *(c)* Consonants show partial separation by voicing.

Table 3: Context-dependent rhyming accuracy for GPT-4.1. The model achieves perfect accuracy on standard rhyming but shows a strong bias toward primary pronunciations when handling heteronyms. Best results in **bold**.

Test Category	Correct	Accuracy (%)
Standard (easy)	15/15	**100**
Standard (tricky)	5/5	**100**
Heteronym – primary	11/12	92
Heteronym – secondary	7/12	58
Overall heteronym	18/24	75

5 Discussion

5.1 Phonetics vs. Semantics: Different Distributional Patterns

The most striking finding is the qualitative difference between how phonetic and semantic features are distributed across layers. Semantic and syntactic features typically *build* through the network: syntactic information peaks in middle layers, while semantic information strengthens in later layers [Radford et al., 2019]. Phonetic information, by contrast, is strongest at the embedding boundary and gradually degrades.

This makes intuitive sense. Phonetic form is a property of the token *itself*—it does not depend on context (except for heteronyms). The embedding table can encode each token's phonetic profile directly, and the first transformer layer refines these representations slightly before the network shifts its representational capacity toward contextual, semantic features. The progressive compression of phonetic directions in deeper layers (table 2) is the geometric signature of this tradeoff: as the network builds richer semantic representations, it reuses dimensions that previously carried phonetic distinctions.

5.2 Comparison to Prior Work

Our results both confirm and extend the findings of McLaughlin et al. [2025]. We confirm that phonetic information is linearly recoverable from token embeddings, consistent with their approximately 96% probe accuracy. Our lower absolute accuracy (macro F1 = 0.41 vs. their 96%) reflects methodological differences: we use Ridge classifiers with per-phoneme binary predictions and macro-averaged F1 across all 77 phonemes including rare ones, while they use multi-hot logistic probes evaluated differently. The key comparison is qualitative: we show that phonetic information does not *improve* with depth, which their work did not test.

Our finding that articulatory features do not cluster in the general residual stream contrasts with McLaughlin et al. [2025]'s organized vowel chart in phoneme mover head result vectors. This suggests that articulatory organization may be a property of specialized attention heads rather than

of the residual stream as a whole—the general representation encodes phonetic information in a less structured way that specialized circuits then organize for the specific task of rhyme production.

5.3 The Primary-Pronunciation Bias

The heteronym results reveal something about how language models store and access pronunciation. The 92% vs. 58% accuracy gap between primary and secondary pronunciations suggests that embeddings encode a dominant phonetic profile, and contextual processing can sometimes—but not reliably—override it. Words where the model succeeds ("read," "lead," "tear") may have more distinctive syntactic cues distinguishing their pronunciations, while words where it fails ("wind," "bow," "live") have subtler contextual signals.

This connects to our probing finding: if the embedding already commits to a phonetic profile that does not improve with depth, then resolving heteronyms requires the model to *override* embedding-level information using contextual attention—a more demanding computation than simply propagating what the embedding already encodes.

5.4 Limitations

Single-token constraint. We analyze only words that GPT-2 encodes as a single token, biasing our sample toward common, short words. Multi-token words constitute the majority of the vocabulary and may exhibit different phonetic encoding patterns, particularly since their pronunciation must be assembled from subword pieces.

Model and probe choices. Our experiments use GPT-2 MEDIUM, a smaller and older model than the LLAMA-3.2-1B studied by McLaughlin et al. [2025]. The phoneme mover head phenomenon may not exist in GPT-2, or may manifest differently. Our use of Ridge classifiers rather than logistic regression means absolute probe accuracies are not directly comparable across studies.

Isolated token analysis. We extract activations for single tokens without surrounding context, which means we measure what is *stored* in each token's representation rather than what is *computed* via attention from neighboring tokens. This is intentional—we want to isolate the static phonetic content—but it misses the role of attention mechanisms that McLaughlin et al. [2025]'s phoneme mover heads exploit.

Heteronym sample size. Our context-dependent test uses only 12 heteronyms on a single model (GPT-4.1), limiting statistical power. A larger-scale heteronym evaluation across multiple models would strengthen these findings.

Pronunciation ambiguity. We use the first transcription from WIKIPRON for each word. Some words have multiple valid pronunciations, and our probes may partially reflect this ambiguity.

6 Conclusion

We investigated where rhyming knowledge is encoded in transformer language models through three experiments on GPT-2 MEDIUM using 13,105 words with IPA transcriptions. Our layer-wise probing reveals that phonetic information is *front-loaded*: it peaks at the embedding/layer-0 boundary and does not improve through the network, contrasting with semantic and syntactic features that build through deeper layers. Our geometric analysis shows that each phoneme has a distinct direction in embedding space ($5.3\times$ above random baseline) but these directions are not orthogonal—they form an overlapping phonetic manifold that compresses in deeper layers. Our heteronym test uncovers a primary-pronunciation bias in GPT-4.1 (92% primary vs. 58% secondary accuracy), demonstrating that context-dependent phonetic processing remains an open challenge.

The answer to "where is rhyming?" is: mostly in the token embedding table, refined by the first transformer layer, and preserved but not enriched through the rest of the network.

Several directions for future work emerge from these findings. Replicating this analysis on LLAMA-3.2-1B and other model families would test whether the front-loading pattern generalizes. Causal interventions—modifying phoneme direction vectors and observing changes in rhyming behavior—would establish whether these directions are merely correlated with phonetic content or causally implicated. Extending the heteronym evaluation to more words and models would better character-

ize the primary-pronunciation bias and test whether improved prompting or fine-tuning can mitigate it. Finally, analyzing multi-token words would reveal whether the phonetic encoding patterns we observe for single tokens extend to subword-assembled pronunciations.

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